

PayRollEx: Web-Based Payroll & Attendance Management System with BioStar 2 Fingerprint Integration

1. Project Overview

We propose a **web-based payroll and attendance management system** connected with the client's **BioStar 2 fingerprint attendance machine**. This system will manage employee attendance, salary calculation, leave, overtime, allowances, deductions, and payroll approvals through a centralized web portal.

The solution is **not a standalone desktop program**. It will be a secure web application accessible through browser login for HR, Accounts, Management, and employees.

BioStar 2 supports web/API-based integration and time-attendance functions, including punch logs, shift schedules, and attendance-related data through its TA/API features.

2. Main Solution Modules

A. Employee Management

PayrollEx
Payroll & Attendance System

Employee Management

Back to Dashboard

Summary Cards:

- Total Employees: 256 (Active Employees)
- Active Employees: 217 (84.77% of Total)
- New This Month: 8 (New Joiners)
- Departments: 12 (Departments)
- Mapped to BioStar 2: 205 (80.08% of Total)

Employee List

Filters: All Departments, All Status, Search employees... Export Add Employee

EMP ID	PHOTO	FULL NAME	DEPARTMENT	DESIGNATION	EMP TYPE	STATUS	BIOSTAR 2 ID	ACTIONS
EMP001		Nimal Perera	IT Department	Senior Developer	Permanent	Active	1001	
EMP002		Kavindi Silva	HR Department	HR Executive	Permanent	Active	1002	
EMP003		Minura Fernando	Finance Department	Accountant	Permanent	Active	1003	
EMP004		Tharushi De Silva	Marketing Department	Marketing Executive	Contract	Active	1004	
EMP005		Kasun Rajapaksa	Operations Department	Operations Manager	Permanent	Active	1005	
EMP006		Isuri Madushani	IT Department	UI/UX Designer	Contract	Inactive	-	
EMP007		Ravindu Bandara	Finance Department	Assistant Accountant	Permanent	Active	1006	
EMP008		Pavithra Jayasinghe	HR Department	HR Assistant	Permanent	Active	1007	

Showing 1 to 8 of 256 entries

Employee Details: Nimal Perera
Senior Developer
Active
EMP ID: EMP001

Personal Info

- Full Name: Nimal Perera
- NIC / Passport: 199123456789
- Date of Birth: 1991-05-15
- Gender: Male
- Marital Status: Married
- Nationality: Sri Lankan
- Date of Joining: 2022-01-10
- Employment Type: Permanent
- Salary Type: Monthly
- Basic Salary: LKR 150,000.00
- EPF Number: 1234567
- BioStar 2 User ID: 1001

Emergency Contact

- Name: Sunil Perera
- Relationship: Brother
- Contact Number: 077 123 4567

Address

- Address: 123, Galle Road, Colombo 04, Sri Lanka

Edit Employee

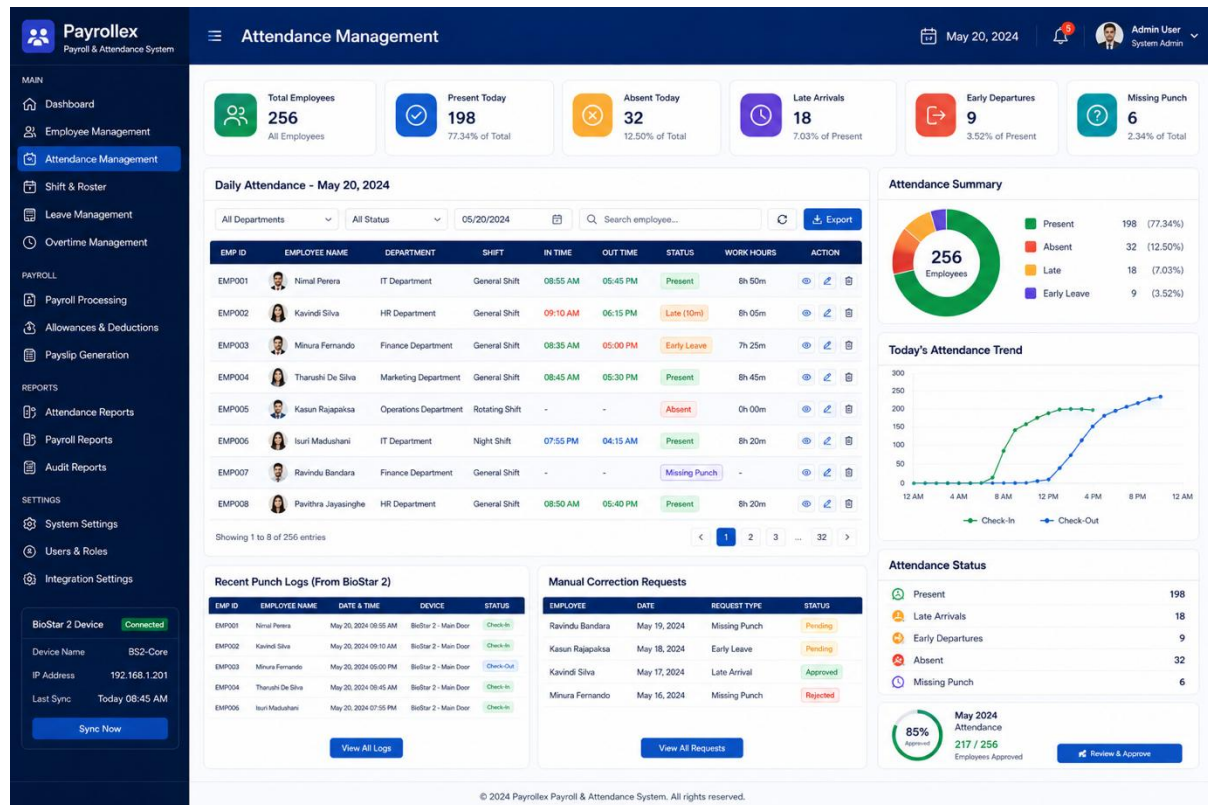
© 2024 PayrollEx System. All rights reserved. Version 1.0.0

The system will maintain complete employee records, including:

Employee ID, name, NIC/passport, department, designation, employment type, salary type, joining date, bank details, EPF number, contact details, emergency contact, and employment status.

Each employee can be mapped with the fingerprint machine user ID, so attendance records from BioStar 2 can be correctly matched with payroll records.

B. Attendance Management



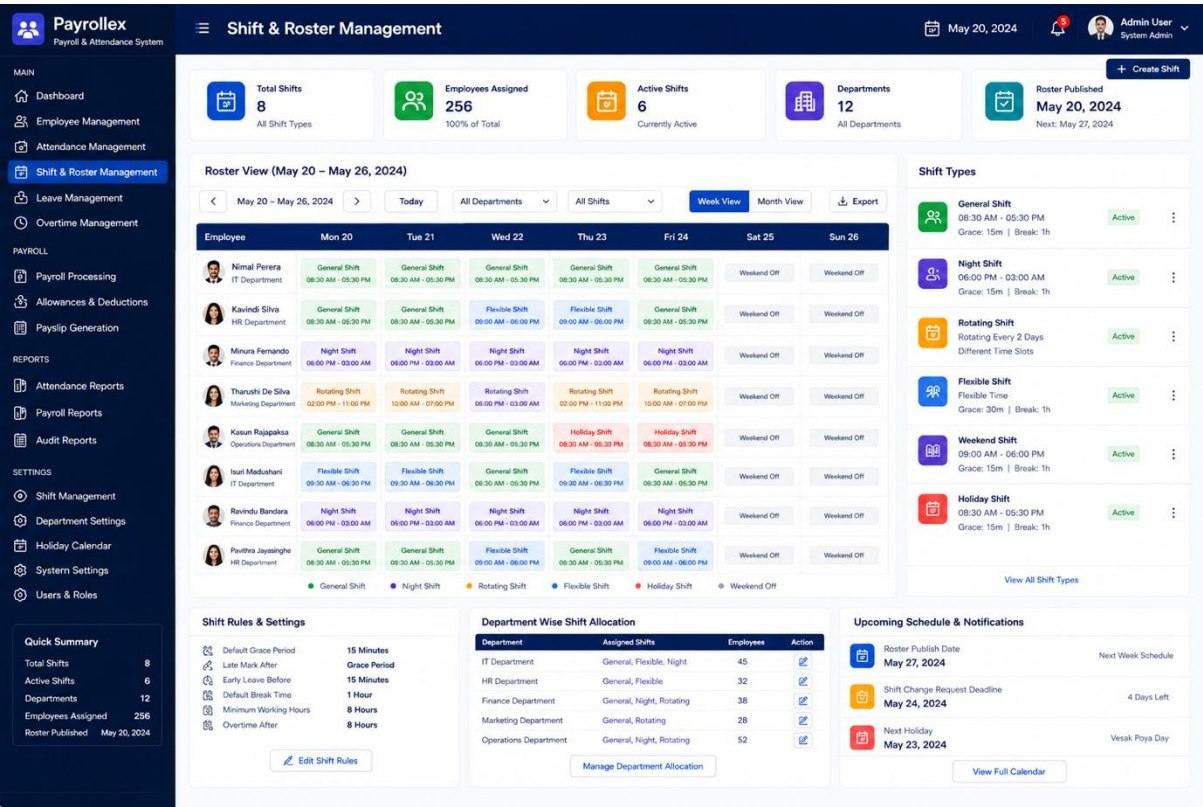
Attendance will be captured from the BioStar 2 fingerprint device and synchronized with the web payroll system.

Features include:

Daily attendance, in-time and out-time, late arrivals, early departures, missing punches, absent days, half-days, shift-based attendance, manual correction requests, and attendance approval.

The system can retrieve attendance/punch logs from BioStar 2 using API-based integration. Suprema documentation confirms BioStar 2 provides API access for retrieving event logs and TA data.

C. Shift and Roster Management

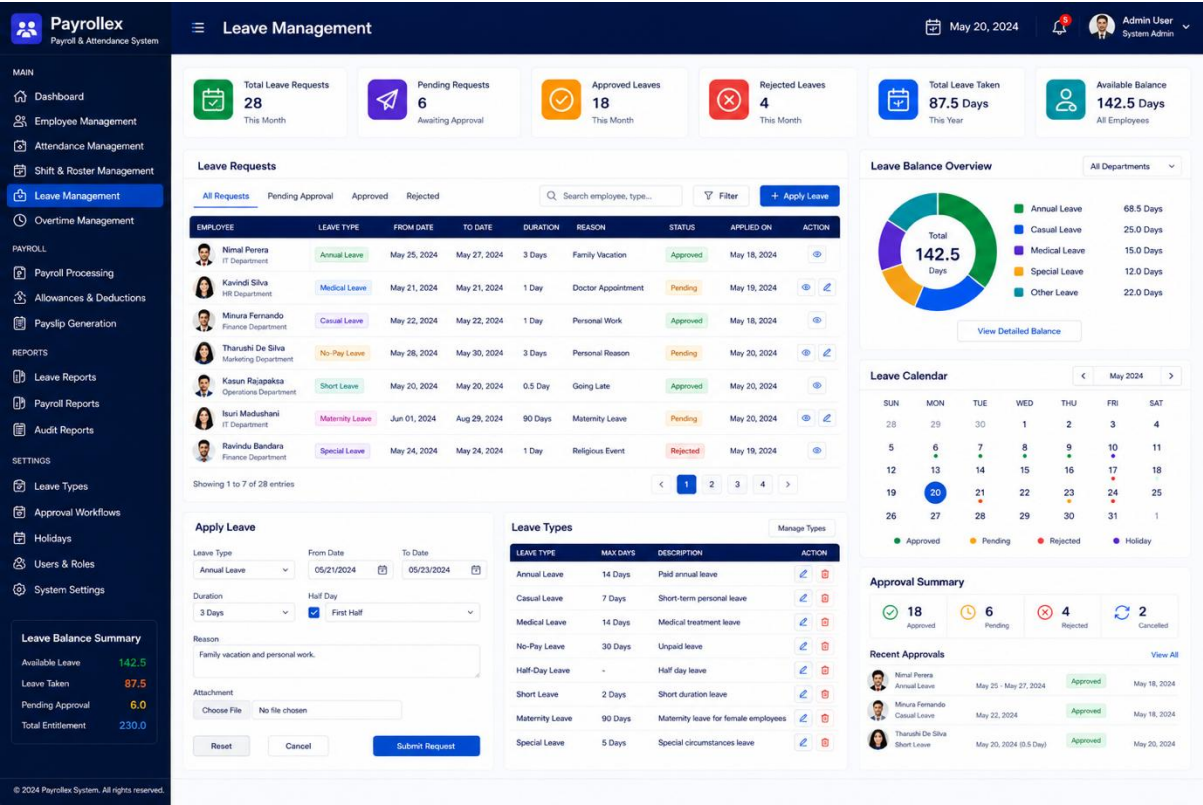


The system will support:

Normal shift, night shift, rotating shift, flexible shift, weekend shift, holiday shift, grace period, lunch break rules, and department-wise shift allocation.

HR can assign shifts to employees, and attendance will be calculated according to the assigned shift rules.

D. Leave Management



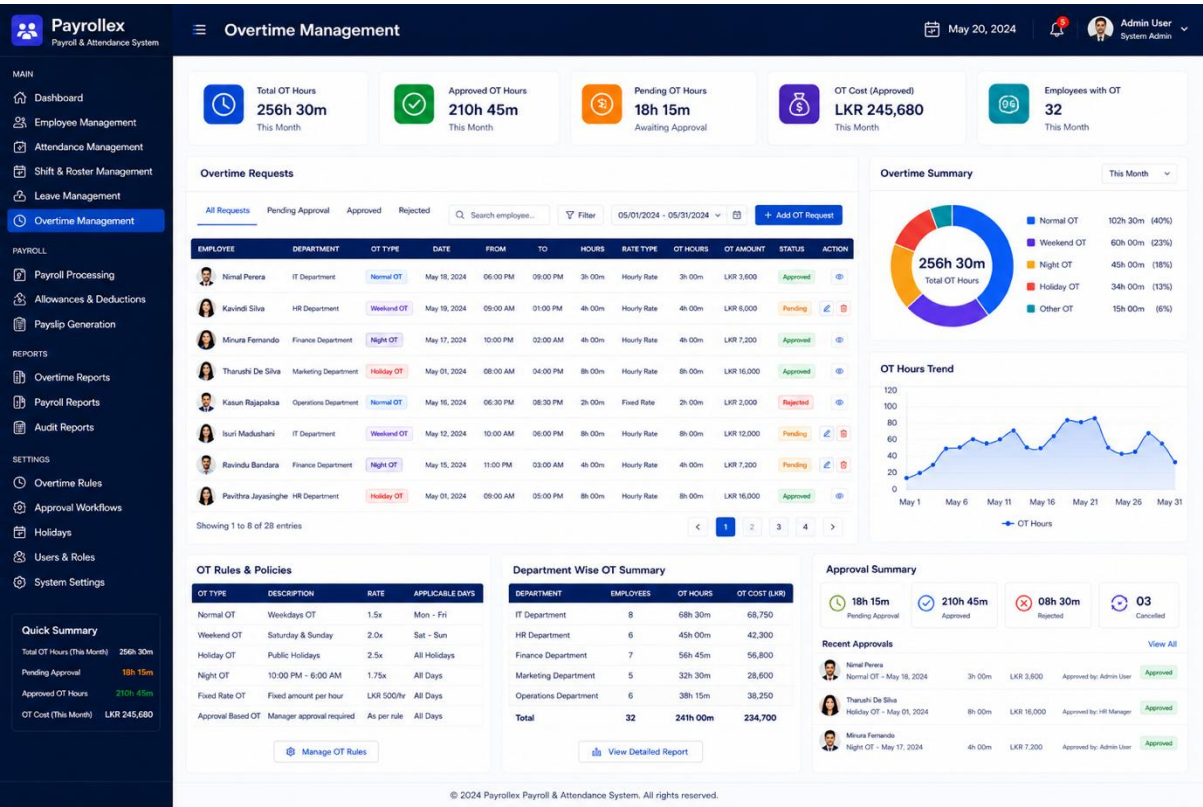
Employees can apply for leave through the web portal.

Leave types include:

Annual leave, casual leave, medical leave, no-pay leave, half-day leave, short leave, maternity leave, and special leave.

Managers can approve or reject leave requests. Approved leave will automatically affect payroll calculation.

E. Overtime Management



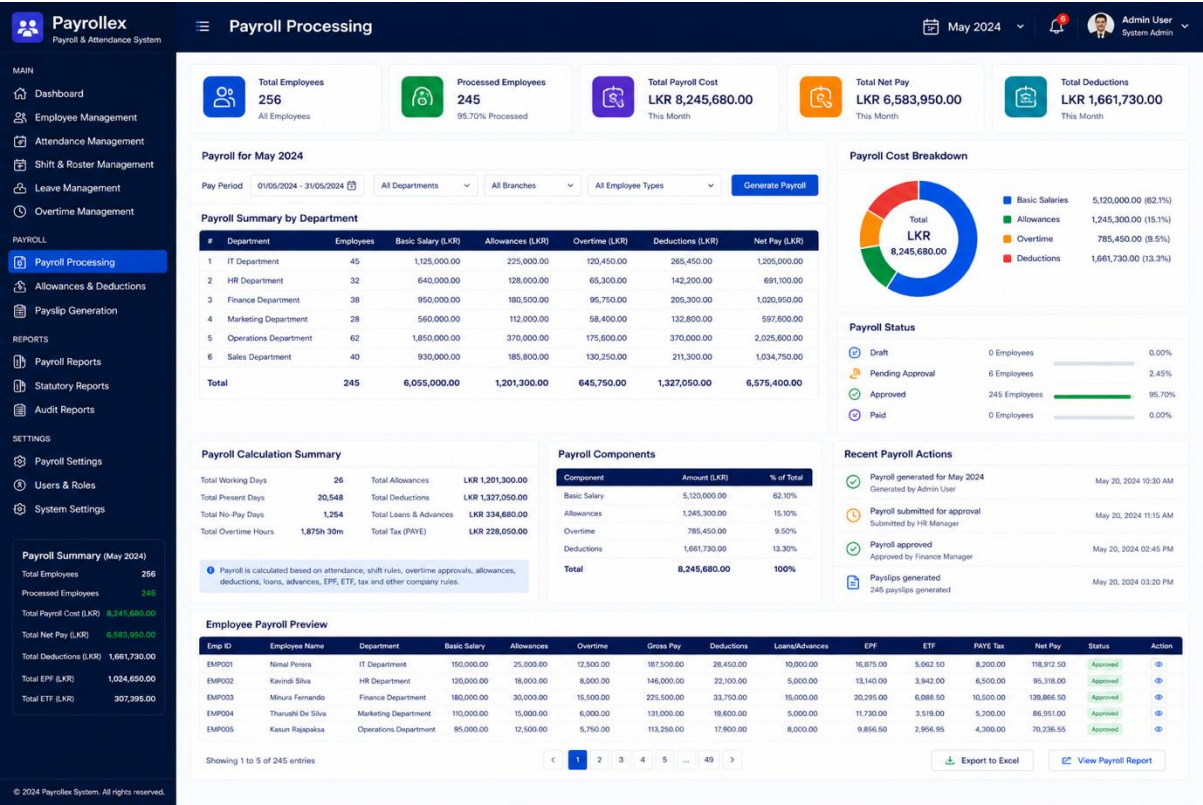
The system will calculate overtime based on company policy.

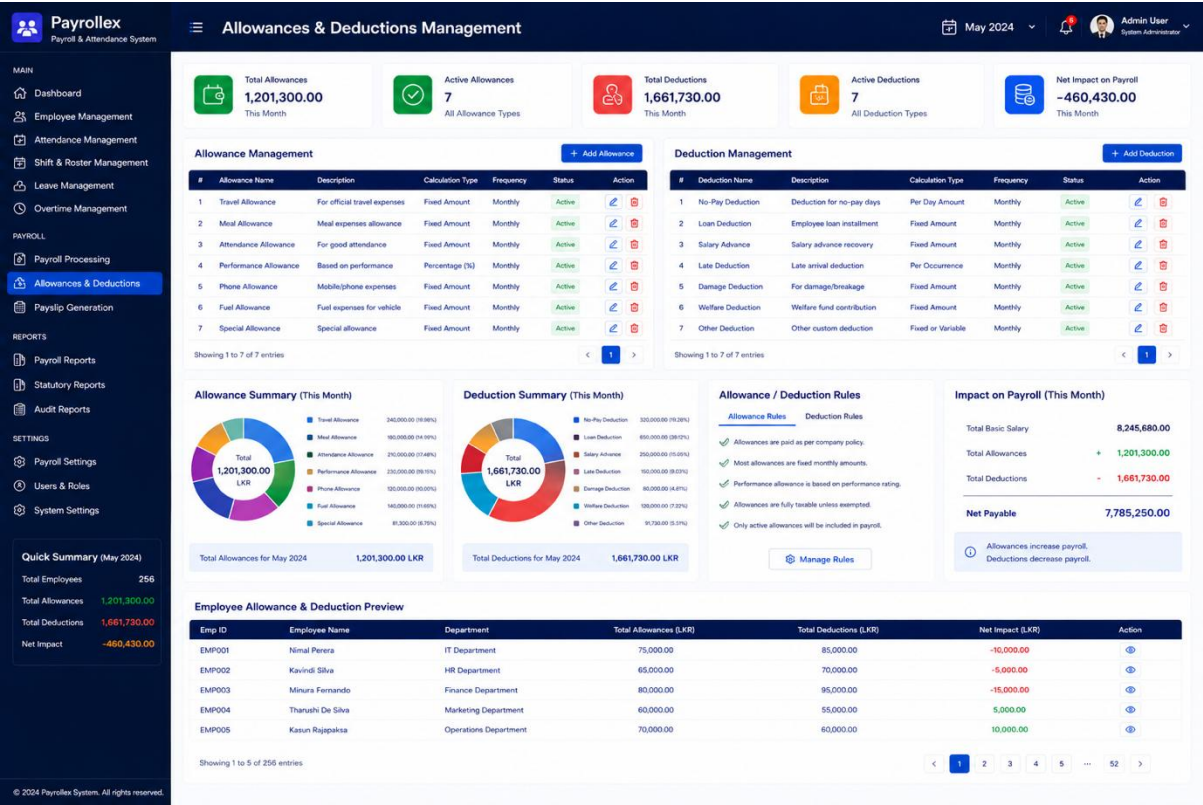
Supported rules include:

Normal OT, weekend OT, holiday OT, night OT, fixed-rate OT, hourly-rate OT, and approval-based OT.

Only approved overtime will be added to payroll.

F. Payroll Processing





Allowance examples:

Travel allowance, meal allowance, attendance allowance, performance allowance, phone allowance, fuel allowance, and special allowance.

Deduction examples:

No-pay deduction, loan deduction, salary advance, late deduction, damage deduction, welfare deduction, and other custom deductions.

I. Payslip Generation

PayrollEx

Payroll & HR System

MAIN

Dashboard

Employee Management

Attendance Management

Shift & Roster Management

Leave Management

Overtime Management

PAYROLL

Payroll Processing

Allowances & Deductions

Payslip Generation

REPORTS

Payroll Reports

Statutory Reports

Audit Reports

SETTINGS

Payroll Settings

Users & Roles

System Settings

Quick Summary (May 2024)

Total Employees

256

Payslips Generated

245

Pending Payslips

11

Total Payroll Cost

LKR 8,245,690.00

Payslip Management

May 2024

Admin User

HR Manager

Pay Period

May 2024 (01 May 2024 - 31 May 2024)

Department

All Departments

Employee

Nimal Perera (EMP001)

View Payslip

Email Payslip

Download PDF

ABC (Pvt) Ltd

123, Business Park, Colombo 05, Sri Lanka

Tel: 011 2 345 678 | Email: info@abc.lk

Reg No: PV 123456

Payslip - May 2024

CONFIDENTIAL

Employee Details

Employee ID

EMP001

Employee Name

Nimal Perera

Designation

Software Engineer

Department

IT Department

Date of Joining

15 Jan 2022

Bank Name

Commercial Bank

Bank Account No.

1234 5678 9012

Salary Details

Pay Period

01 May 2024 - 31 May 2024

Payment Date

31 May 2024

Payment Mode

Bank Transfer

Employee Type

Permanent

Attendance Summary

Total Working Days

26

Present Days

24

Absent Days

1

Late Days

1

Half Days

0

Paid Leave Days

0

No-Pay Days

0

Total Overtime Hours

12h 30m

Earnings (Allowances)

Basic Salary

150,000.00

Travel Allowance

15,000.00

Meal Allowance

10,000.00

Performance Allowance

20,000.00

Phone Allowance

8,000.00

Fuel Allowance

12,000.00

Special Allowance

5,000.00

Total Allowances

220,000.00

Amount (LKR)

150,000.00

15,000.00

10,000.00

20,000.00

8,000.00

12,000.00

5,000.00

220,000.00

Deductions

No-Pay Deduction

0.00

Loan Deduction

10,000.00

Salary Advance

0.00

Late Deduction

500.00

Damage Deduction

0.00

Welfare Deduction

500.00

Other Deductions

0.00

Total Deductions

11,000.00

Amount (LKR)

0.00

10,000.00

0.00

500.00

0.00

500.00

0.00

11,000.00

Statutory Deductions

EPF (Employee 8%)

12,000.00

ETF (Employee 3%)

4,500.00

Total Statutory Deductions

16,500.00

Amount (LKR)

12,000.00

4,500.00

16,500.00

Gross Pay (Earnings)

370,000.00

Total Deductions

27,500.00

Net Pay

342,500.00

Net Pay in Words: Three Hundred Forty Two Thousand Five Hundred Rupees Only.

This is a computer generated payslip and does not require a signature.

Thank you for your contribution!

Payslip History

May 2024

Generated

Apr 2024

Generated

Mar 2024

Generated

Feb 2024

Generated

Jan 2024

Generated

Dec 2023

Generated

Payslip Delivery

Email

nimal.perera@example.com

Sent

Download PDF

Available

Payslips are generated based on approved attendance, leaves, overtime, allowances and deductions.

The system will generate monthly payslips with:

Employee details, salary period, basic salary, attendance summary, allowances, deductions, EPF/ETF, net salary, and company details.

Payslips can be downloaded as PDF or emailed to employees.

J. Reports and Dashboards

Reports include:

Monthly payroll report, attendance report, late report, absent report, overtime report, leave report, EPF/ETF report, department salary report, bank transfer report, payslip report, and audit report.

Management dashboard will show total employees, present/absent count, monthly payroll cost, overtime cost, leave summary, and pending approvals.

3. BioStar 2 Fingerprint Integration

BioStar 2 Fingerprint Integration

May 20, 2024 | Admin User (System Administrator)

Integration Status: Active (All Systems Operational)

Integration Flow:

- Employee Registered:** Employee is registered in BioStar 2 fingerprint machine.
- Employee Mapping:** Employee fingerprint / user ID is mapped with payroll employee ID.
- PUNCH:** BioStar 2 records punch-in and punch-out events.
- Data Synchronization:** Web payroll system synchronizes attendance logs via API.
- Attendance Calculation:** System calculates daily attendance based on shift rules.
- HR Verification:** HR verifies and approves attendance records.
- Payroll Generation:** Payroll is generated from approved attendance data.

Integration Architecture:

BioStar 2 System: Fingerprint Devices (Suprema Devices), User Management (Employees), Event Logs (Punch In/Out), Time Attendance Data.

Web Payroll System: Employee Mapping (BioStar User ID ↔ Payroll ID), Attendance Logs Synchronization, Attendance Processing (Shift Rules & Calculation), Reports & Payroll Generation.

API Communication: Web API (HTTPS), JSON Data Format, Secure Authentication, Real-time / Scheduled Sync.

Recent Integration Logs:

Time	Event	Details	Status
May 20, 2024 10:15 AM	Attendance Sync	1,245 records imported	Success
May 20, 2024 10:00 AM	Device Connection	Main Office Device	Success
May 20, 2024 09:45 AM	Attendance Sync	1,198 records imported	Success
May 20, 2024 09:30 AM	Device Connection	Branch Office Device	Success
May 20, 2024 09:15 AM	Attendance Sync	1,150 records imported	Success

About BioStar 2

BioStar 2 by Suprema is a web-based open platform that provides powerful security and time attendance features with an open API/SDK for easy integration with third-party systems.

- Web-based Open Platform
- Supports Time Attendance
- Web API / SDK Support
- Real-time Device Communication

Integration Benefits:

- Accurate attendance data capture from fingerprint devices
- Real-time synchronization reduces manual work
- Eliminates data entry errors
- Efficient payroll processing with approved attendance
- Secure and reliable API-based integration

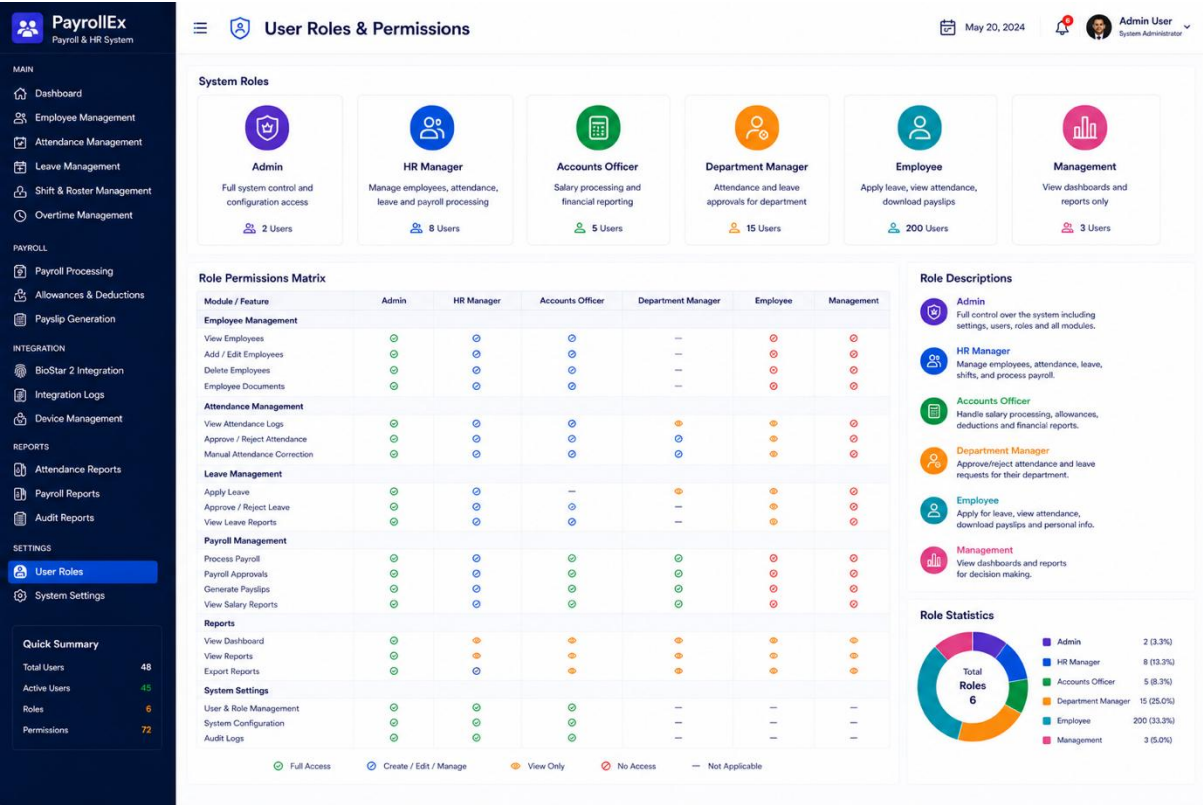
The proposed system will connect with BioStar 2 through API integration.

Integration Flow

1. Employee is registered in BioStar 2 fingerprint machine.
2. Employee fingerprint/user ID is mapped with payroll employee ID.
3. BioStar 2 records punch-in and punch-out events.
4. Web payroll system synchronizes attendance logs.
5. System calculates daily attendance based on shift rules.
6. HR verifies attendance.
7. Payroll is generated from approved attendance data.

BioStar 2 is described by Suprema as a web-based open platform supporting time attendance and third-party integration through web APIs/SDKs.

4. User Roles



The system will include role-based access:

Admin: full system control
HR Manager: employee, attendance, leave, payroll processing
Accounts Officer: salary processing and financial reports
Department Manager: attendance and leave approvals
Employee: leave application, attendance view, payslip download
Management: dashboard and reports only

5. Security Features

Security will include:

Secure login, role-based permission, HTTPS hosting, encrypted passwords, audit logs, database backup, access control, payroll approval workflow, and restricted salary visibility.

6. Web-Based System Advantages

Because this is a web-based system:

It can be accessed from multiple branches, no installation is required on every computer, updates are centralized, reports are available in real time, management can access dashboards remotely, and the system can easily integrate with future modules such as HRM, accounting, inventory, or ERP.